

A classification of finite Euclidean Ramsey configurations

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Abstract

We classify finite Euclidean Ramsey configurations by a necessary and sufficient tensor condition over their coordinate fields. The Ramsey property here concerns monochromatic congruent copies at the original scale under arbitrary finite colorings. The criterion shows that every nonempty subtransitive set and every nonempty set of at most five points on a circle is Ramsey. In particular, some Ramsey cyclic quadrilaterals are not subtransitive, disproving the necessity direction of the Leader–Russell–Walters conjectured characterization.

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1 The classification problem

A finite nonempty set A in a Euclidean space is *Ramsey* if, for every integer $r \geq 2$, there is an integer $D \geq 1$ such that every map $c : \mathbb{R}^D \rightarrow \{1, \dots, r\}$ is constant on a congruent copy of A .

Congruence means preservation of every pairwise distance, at the original scale. The coloring is an arbitrary function. In particular, no measurability or other regularity is assumed.

Erdős, Graham, Montgomery, Rothschild, Spencer and Straus introduced Euclidean Ramsey theory and proved that every finite Ramsey set is spherical [4, Theorem 13]. Graham later conjectured that sphericity is also sufficient [7]. Major positive results established the Ramsey property for nondegenerate triangles and then all nondegenerate simplices, by Frankl and Rödl [5, 6]; finite sets with a soluble transitive isometry group and cyclic trapezoids, by Kříž [13, 14]; and vertex sets of regular polytopes, by Cantwell [2].

Leader, Russell and Walters proposed a different characterization: a finite set is Ramsey exactly when it embeds in a finite transitive Euclidean set [16, Conjecture A]. Transitivity means that the isometries of the finite set act transitively on its points. An isometric subset of such a set is called *subtransitive*. Their group-theoretic and Hales–Jewett formulations led to the Block Sets Conjecture; subsequent work studies the possible block sizes [10]. The link with soluble symmetry has also expanded: Karamanlis embedded every simplex in a finite product of regular polygons [12], and Behague constructed soluble transitive extensions for many known Ramsey classes and further permutation configurations [1]. Ivan, Leader and Walters developed generalized-prism constructions, with Ramsey conclusions under a soluble symmetry hypothesis [11]. Moore and, independently, Mirabi proved that adjoining a point outside the affine span of a Ramsey set preserves the Ramsey property [19, 18]. The affine-span restriction leaves the case of arbitrary coplanar circle points beyond this extension theorem.

Pálvölgyi has recently given a seven-point circle configuration that is not Ramsey, disproving the spherical conjecture [20, Theorem 1]. The appendix to that version also states that generic seven-point circle configurations are not Ramsey. Its author’s note announces a proof that all finite transitive sets are Ramsey and says that its exposition is still being prepared.¹ Our main theorem gives a necessary and sufficient algebraic condition for an arbitrary finite Euclidean configuration.

1.1 The exact tensor criterion

The problem is to characterize these finite sets up to congruence. A singleton is Ramsey. For any other set, let $d \geq 1$ be its affine dimension, and choose a congruent representative $A = \{a_1, \dots, a_s\} \subset \mathbb{R}^d$ whose affine span is $\mathbb{R}^d$. The points are distinct and $s \geq 2$. Define

$$p_i = \begin{pmatrix} 1 \\ a_i \end{pmatrix}, \quad F = \mathbb{Q}((a_i)_\alpha : 1 \leq i \leq s, 1 \leq \alpha \leq d), \quad B = F \otimes_{\mathbb{Q}} F.$$

The commutative ring B has a multiplication homomorphism

$$m_F : B \longrightarrow F, \quad m_F(x \otimes y) = xy.$$

An equality in B is stronger than the equality obtained by applying m_F ; this map need not be injective. Index the entries of p_i and the rows and columns of a $(d+1)$ -by- $(d+1)$ matrix by $0, 1, \dots, d$. Thus index 0 is the constant coordinate.

Theorem 1.1 (Classification). *Let $A = \{a_1, \dots, a_s\} \subset \mathbb{R}^d$ be a finite set of at least two points with affine span $\mathbb{R}^d$, and define p_i, F, B, m_F as above. Then A is Ramsey if and only if there is a matrix $P \in \text{Mat}_{d+1}(B)$ such that*

$$(p_i \otimes 1)^T P (1 \otimes p_i) = 0 \quad (1 \leq i \leq s), \quad (1)$$

$$m_F(P_{\alpha\beta}) = \delta_{\alpha\beta} \quad (1 \leq \alpha, \beta \leq d). \quad (2)$$

¹Pálvölgyi attributes the writing, proofs, and ideas in that manuscript to ChatGPT. He labels the additional appendix claims as unchecked by him.

Here the entries of $p_i \otimes 1$ and $1 \otimes p_i$ are formed componentwise. There are no separate conditions on the multiplied constant row or constant column of P .

The condition is independent of the chosen congruent representative in its affine dimension; we also prove this directly through its tensor formulation. It uses exact relations in the coordinate field. It is not a procedure for recovering those relations from numerical approximations to arbitrary real coordinates.

1.2 Geometric consequences

Applying m_F to the evaluation equations gives an ordinary sphere equation

$$\|a_i\|^2 + \ell \cdot a_i + c = 0 \quad (1 \leq i \leq s)$$

for some $\ell \in F^d$ and $c \in F$. The tensor condition requires the evaluation identities to hold before multiplication. This is the distinction between sphericity and the Ramsey property that the classification detects.

Every subtransitive set satisfies the tensor condition. Another sufficient condition is that evaluation at the points of a spherical set gives independent functionals on the space of quadratic polynomials. It applies to every nonempty set of at most five points on a circle, and hence to every cyclic quadrilateral. In Corollary 7.5, we combine this result with the non-subtransitive cyclic kites of Leader, Russell and Walters [15, Corollary 2]. Those kites are Ramsey, disproving the necessity direction of their proposed characterization and their non-Ramsey conjecture for the same family [15, Conjecture 3].

For negative examples, nine points obtained from algebraically independent real parameters in a rational parametrization of a circle fail the tensor condition. Pálvölgyi's generic seven-point claim [20, Theorem A.3] already implies this non-Ramsey conclusion; our determinant calculation illustrates how to apply the classification directly. We also give a tensor obstruction for a twelve-point configuration consisting of three rotated coordinate squares, using the same type of weighted derivation moments as his heptagon construction. These consequences are proved after the classification and require no further Ramsey construction.

1.3 How the algebra forces a monochromatic copy

The proof begins by replacing the coordinate-field matrix with a finite rational combination of tensors of affine functions on $\mathbb{R}^d$. Its evaluations at all points of A vanish as rational tensors, while its gradient Gram matrix is I_d . If no such tensor exists, algebraic separation produces a finite coloring that avoids A in every dimension. This proves necessity.

For sufficiency, the difficulty is to convert the tensor cancellations into a congruent copy at scale one: arbitrary colorings provide no continuity with which to absorb approximation errors. The proof has three stages.

First, the tensor determines signed rational multiplicities for a finite list of affine functions. At every point of A , the positive and negative multiplicities cancel separately at each evaluation value. A Gaussian construction, followed by discretization and rational approximation within a fixed convolution image, makes the negative gradient second moment small while retaining these cancellations exactly. The positive and negative multiplicities then give two lists of affine functions. Evaluating the lists at a_i gives vectors f_i, g_i with

$$\|f_i - f_j\|^2 = \|a_i - a_j\|^2, \quad \|g_i - g_j\|^2 = q\|a_i - a_j\|^2,$$

where $q > 0$ can be arbitrarily small, and each g_i is a coordinate permutation of f_i . Both the distances and the matching coordinate multisets are exact.

Second, we associate a free-group generator to each real sequence with finite support. A *path* is an ordered product of s -tuples: each factor either records an ordered scaled copy of A or has the same group element in every position. Its *endpoint* is the coordinatewise product, and its *weight* is the sum of the squared copy scales. We prove that equal endpoints with arbitrarily small weight ratios force the Ramsey property. The obstacle, familiar from the group and word methods of Leader, Russell and Walters [16, Sections 2–3], is the initially unknown number t of variable positions in a Hales–Jewett line. After fixing the Hales–Jewett word length n , we prepare one endpoint at weights $1, 1/2, \dots, 1/n$ before coloring the words. An ultrafilter construction then realizes the products by finite strings, keeping common factors identical. The t variable positions give squared-distance multiplier $t(1/t) = 1$. Compactness supplies a finite ambient dimension for each number of colors.

Third, the permutation pairs from the first stage must be made to have equal path endpoints. Permutation averaging puts the discrepancies between their group entries in a subgroup whose elements can be corrected by appending paths of unit-scale copies. Further averaging makes the total correction weight small relative to the larger original weight. Appending these corrections produces the equal endpoints and arbitrarily small weight ratios required in the second stage.

Sections 2–4 establish the algebraic formulation, necessity, and the exact pair construction. Section 5 states and proves the group-endpoint criterion that suffices for Ramsey forcing. Section 6 constructs the required endpoints and completes the proof of Theorem 1.1. Section 7 derives the geometric examples. All arguments take place in ZFC.

2 Rational tensors and the coloring obstruction

We first express the matrix condition of Theorem 1.1 as a condition on affine functions. This formulation makes the obstruction to arbitrary colorings explicit and supplies the input for the construction in the next section. Throughout, $s \geq 2$, $d \geq 1$, and $A = \{a_1, \dots, a_s\}$ affinely spans $\mathbb{R}^d$; the field F , ring B , columns p_i , and multiplication map m_F are as in the theorem.

Put $W = \mathbb{R} \times \mathbb{R}^d$, regarded as a vector space over $\mathbb{Q}$. An element $w = (v, u)$ specifies the affine function $a \mapsto v + a \cdot u$. For each label define the $\mathbb{Q}$ -linear evaluation map

$$e_i : W \longrightarrow \mathbb{R}, \quad e_i(v, u) = v + a_i \cdot u.$$

Let $\mathcal{T}(W)$ be the subspace of $W \otimes_{\mathbb{Q}} W$ fixed by the flip $w \otimes w' \mapsto w' \otimes w$. Thus our symmetric tensors are flip-invariant tensors. The map

$$(v, u) \otimes (v', u') \longmapsto u(u')^{\top}$$

is $\mathbb{Q}$ -linear on the tensor product and restricts to a map $G : \mathcal{T}(W) \rightarrow \text{Sym}_d(\mathbb{R})$, where $\text{Sym}_d(\mathbb{R})$ denotes the real symmetric matrices. In particular, $G(w \otimes w) = uu^{\top}$. The tensor condition is the existence of $T \in \mathcal{T}(W)$ such that

$$(e_i \otimes e_i)(T) = 0 \quad (1 \leq i \leq s), \quad G(T) = I_d. \quad (3)$$

The first identities hold in $\mathbb{R} \otimes_{\mathbb{Q}} \mathbb{R}$. They are stronger than their images under real multiplication.

Proposition 2.1. *The matrix condition in Theorem 1.1 is equivalent to (3).*

Proof. Write $\mathcal{B} = \mathbb{R} \otimes_{\mathbb{Q}} \mathbb{R}$, let $\tau(x \otimes y) = y \otimes x$, and let $m_{\mathbb{R}} : \mathcal{B} \rightarrow \mathbb{R}$ be multiplication. The coordinate decomposition of W identifies $W \otimes_{\mathbb{Q}} W$ with arrays $Q = (Q_{\alpha\beta})_{0 \leq \alpha, \beta \leq d}$ over $\mathcal{B}$. Under this identification, evaluation is

$$E_i(Q) = \sum_{\alpha, \beta=0}^d (p_{i,\alpha} \otimes 1) Q_{\alpha\beta} (1 \otimes p_{i,\beta}),$$

and the gradient matrix is the spatial block $(m_{\mathbb{R}}(Q_{\alpha\beta}))_{1 \leq \alpha, \beta \leq d}$. Tensor flip becomes

$$Q_{\alpha\beta}^* = \tau(Q_{\beta\alpha}).$$

Directly from the evaluation formula, $E_i(Q^*) = \tau(E_i(Q))$, and multiplication of Q^* transposes the multiplied array. Consequently $\frac{1}{2}(Q + Q^*)$ preserves every zero evaluation and any multiplied spatial block equal to I_d . This proves the implication from a matrix over B to a symmetric tensor, using the natural inclusion $B \subset \mathcal{B}$.

For the converse, we show that extending the coefficient ring from B to $\mathcal{B}$ introduces no new solutions. Choose an F -basis $(r_\lambda)_{\lambda \in \Lambda}$ of $\mathbb{R}$. Distributing tensor products over direct sums gives the free B -module decomposition

$$\mathcal{B} = \bigoplus_{\lambda, \mu \in \Lambda} B(r_\lambda \otimes r_\mu). \quad (4)$$

Here a coefficient $x \otimes y \in B$ in the indicated summand represents $(xr_\lambda) \otimes (yr_\mu)$. If an array Q over $\mathcal{B}$ solves the evaluation equations and has multiplied spatial block I_d , its entries have a common finite expansion

$$Q = \sum_{(\lambda, \mu) \in J} (r_\lambda \otimes r_\mu) P^{\lambda\mu}, \quad P^{\lambda\mu} \in \text{Mat}_{d+1}(B).$$

All coefficients in the maps E_i belong to B . Thus (4) implies $E_i(P^{\lambda\mu}) = 0$ for each pair and each label. Let $C^{\lambda\mu} \in \text{Mat}_d(F)$ be the multiplied spatial block of $P^{\lambda\mu}$. Applying multiplication to the expansion gives

$$I_d = \sum_{(\lambda, \mu) \in J} r_\lambda r_\mu C^{\lambda\mu}.$$

This finite linear system has coefficients and right-hand side in F . Since it is consistent over $\mathbb{R}$, row reduction over F shows that it is consistent over F . Choose $c_{\lambda\mu} \in F$ with $I_d = \sum c_{\lambda\mu} C^{\lambda\mu}$. Then

$$P = \sum_{(\lambda, \mu) \in J} (c_{\lambda\mu} \otimes 1) P^{\lambda\mu}$$

is a solution over B . The individual matrices $C^{\lambda\mu}$ need not be symmetric; the descent took place in the full space $\text{Mat}_d(F)$. If a symmetric tensor is wanted after descent, the symmetrization already proved applies to P . $\square$

Remark 2.2. The finite coordinate presentation explains the algebraic nature of the criterion. Let $F_0 = \mathbb{Q}[(a_i)_j]$, the subring of $\mathbb{R}$ generated by all coordinates. It is a finitely generated domain and $F = \text{Frac}(F_0)$. The ring B is the localization of $F \otimes_{\mathbb{Q}} F_0$ at the elements $1 \otimes f$ for $0 \neq f \in F_0$. The latter ring is a finitely generated F -algebra, so B is Noetherian. The kernel of the finite evaluation system therefore has finitely many B -module generators. Its multiplied spatial image is the F -span of their multiplied spatial blocks, because every scalar $c \in F$ lifts to $c \otimes 1$. This description uses exact coordinate-field relations; it makes no claim to recover those relations from numerical real inputs.

Remark 2.3. The tensor condition is directly invariant under congruence. If $a'_i = Qa_i + t$ with $Q \in O(d)$, the invertible $\mathbb{Q}$ -linear map

$$H : W \longrightarrow W, \quad H(v, u) = (v - t \cdot Qu, Qu)$$

satisfies $e'_i \circ H = e_i$ and $G((H \otimes H)T) = QG(T)Q^\top$. It therefore carries certificates for A to certificates for A' ; the inverse congruence gives the converse. Reordering labels merely reorders the equations. Any two congruent representatives in their affine dimension are related by such a map, so Proposition 2.1 also proves representative independence for the matrix criterion, even when their coordinate fields differ.

We next prove the necessary direction of the classification. Failure of (3) gives an algebraic functional, and its values provide a fixed finite coloring that works in every ambient dimension. The invariant-and-coloring strategy goes back to Erdős et al. [4, Theorem 13 and Lemma 15]; here rational tensor separation supplies the invariant, and the modular coloring is proved directly. Their real-variable coloring lemma extends Rado's work on partition regularity [21].

Proposition 2.4. *If (3) has no solution, then there is an integer $r_0 \geq 2$ such that every $\mathbb{R}^D$ admits an r_0 -coloring with no monochromatic congruent copy of A . In particular, every Euclidean Ramsey set satisfies the matrix condition of Theorem 1.1.*

Proof. Regard

$$\mathcal{Y} = (\mathbb{R} \otimes_{\mathbb{Q}} \mathbb{R})^s \oplus \text{Sym}_d(\mathbb{R})$$

as a $\mathbb{Q}$ -vector space and define the $\mathbb{Q}$ -linear map

$$\Phi : \mathcal{T}(W) \longrightarrow \mathcal{Y}, \quad \Phi(T) = ((e_i \otimes e_i)(T))_{i=1}^s \oplus G(T).$$

By assumption $y_0 = (0, \dots, 0, I_d)$ is outside its image. In the quotient vector space $\mathcal{Y}/\text{im } \Phi$, extend the nonzero class of y_0 to a basis over $\mathbb{Q}$. Sending that basis element to 1 and the others to 0 gives a $\mathbb{Q}$ -linear functional $\ell : \mathcal{Y} \rightarrow \mathbb{R}$ such that

$$\ell \circ \Phi = 0, \quad \ell(y_0) = 1.$$

Write its components as $\ell_i : \mathbb{R} \otimes_{\mathbb{Q}} \mathbb{R} \rightarrow \mathbb{R}$ and $L : \text{Sym}_d(\mathbb{R}) \rightarrow \mathbb{R}$, and put $h_i(x) = \ell_i(x \otimes x)$. Applying ℓ to $\Phi(w \otimes w)$ gives

$$\sum_{i=1}^s h_i(v + a_i \cdot u) + L(uu^T) = 0, \quad L(I_d) = 1. \quad (5)$$

Taking $u = 0$ shows that

$$\sum_{i=1}^s h_i(x) = 0 \quad (x \in \mathbb{R}). \quad (6)$$

These are purely algebraic functionals; no continuity or measurability is asserted or needed.

Fix D and define $H_i(z) = \sum_{j=1}^D h_i(z_j)$ for $z \in \mathbb{R}^D$. We claim that every ordered congruent copy $(b_1, \dots, b_s)$ satisfies

$$\sum_{i=1}^s H_i(b_i) = -1. \quad (7)$$

Indeed, equality of the pairwise distances gives equality of the Gram matrices of $(a_i - a_1)_i$ and $(b_i - b_1)_i$ by polarization. Thus sending each $a_i - a_1$ to $b_i - b_1$ defines an inner-product-preserving linear map on their span: any relation among the first differences has squared norm zero for the corresponding combination of the second. Since A affinely spans $\mathbb{R}^d$, there are a linear isometry $Q : \mathbb{R}^d \rightarrow \mathbb{R}^D$ and $t \in \mathbb{R}^D$ such that $b_i = t + Qa_i$. Writing u_j^T for the j th row of Q , we have

$$(b_i)_j = e_i(t_j, u_j), \quad \sum_{j=1}^D u_j u_j^T = Q^T Q = I_d.$$

Sum (5) over j . The finite additivity of L proves (7). Also (6) gives $\sum_i H_i(z) = 0$ at every single point z .

Choose $K = 2s+1$ and partition $[0, 2)$ into K half-open intervals of equal length $2/K < 1/s$. Color a point z by the s -tuple of interval labels of the representatives of $H_i(z)$ modulo 2.

This uses at most $r_0 = K^s$ colors, independently of D . If $b_1, \dots, b_s$ were monochromatic, there would be integers n_i and real numbers ε_i , with $|\varepsilon_i| < 2/K$, such that

$$H_i(b_i) - H_i(b_1) = 2n_i + \varepsilon_i.$$

Summation yields

$$-1 = 2 \sum_i n_i + \sum_i \varepsilon_i, \quad \left| \sum_i \varepsilon_i \right| < 1.$$

The first identity makes the error sum an odd integer, contradicting the second. This coloring avoids A in every dimension and completes the proof. $\square$

The remaining direction begins with a tensor satisfying (3). Its evaluations vanish before multiplication, and that exact cancellation will produce equal coordinate histograms for two embeddings at different scales.

3 A finite lattice identity

Assume that the tensor condition holds. We begin the sufficiency proof by turning its rational tensor equalities into exact cancellation identities for finitely many affine functions. These identities will let us construct two copies of the configuration at different scales whose corresponding points have the same coordinates in different orders.

Recall that $W = \mathbb{R} \times \mathbb{R}^d$ is viewed over $\mathbb{Q}$, and that $e_i(v, u) = v + a_i \cdot u$. Let $T \in W \otimes_{\mathbb{Q}} W$ be symmetric, with

$$(e_i \otimes e_i)T = 0 \quad (1 \leq i \leq s), \quad G(T) = I_d.$$

A tensor uses only finitely many vectors. Choose a basis $w_1, \dots, w_k$ over $\mathbb{Q}$ for a finite-dimensional subspace containing its factors, and write

$$T = \sum_{a,b=1}^k C_{ab} w_a \otimes w_b, \quad C \in \text{Mat}_k(\mathbb{Q}), \quad C = C^T.$$

For $\lambda \in \mathbb{Q}^k$, put

$$w(\lambda) = \sum_{j=1}^k \lambda_j w_j.$$

Write $w_j = (v_j, u_j)$ and let $U = (u_1 \cdots u_k)$ be the real $d \times k$ matrix whose columns are their gradients. The gradient identity for T is

$$UCU^T = I_d. \tag{8}$$

Set $\Gamma = \mathbb{Z}^k$ and define the evaluation lattices

$$\Lambda_i = \{\lambda \in \Gamma : e_i(w(\lambda)) = 0\}, \quad V_i = \text{span}_{\mathbb{Q}} \Lambda_i.$$

Clearing denominators shows that V_i is the kernel of the rational linear map $\lambda \mapsto e_i(w(\lambda))$ on $\mathbb{Q}^k$. Thus $\mathbb{Q}^k/V_i$ embeds into $\mathbb{R}$ over $\mathbb{Q}$. Tensoring this injection with itself remains injective, so the tensor represented by C has zero image in $(\mathbb{Q}^k/V_i) \otimes_{\mathbb{Q}} (\mathbb{Q}^k/V_i)$.

The next lemma converts these tensor equalities into a finitely supported rational function on Γ . Vanishing of its sums on each coset will record exact cancellation among equal evaluations.

Lemma 3.1 (Finite lattice identity). *Let $k, s \geq 1$, let $\Lambda_1, \dots, \Lambda_s$ be subgroups of $\Gamma = \mathbb{Z}^k$, and put $V_i = \text{span}_{\mathbb{Q}} \Lambda_i$. Let $C \in \text{Mat}_k(\mathbb{Q})$ be symmetric. Suppose that the image of $\sum_{a,b} C_{ab} \varepsilon_a \otimes \varepsilon_b$ in $(\mathbb{Q}^k/V_i) \otimes_{\mathbb{Q}} (\mathbb{Q}^k/V_i)$ is zero for every i , where $\varepsilon_1, \dots, \varepsilon_k$ is the standard basis of $\mathbb{Q}^k$. There is a finitely supported function $\nu : \Gamma \rightarrow \mathbb{Q}$ such that*

$$\sum_{\gamma \in q} \nu(\gamma) = 0 \quad (q \in \Gamma/\Lambda_i, \ 1 \leq i \leq s) \quad (9)$$

and

$$\sum_{\gamma} \nu(\gamma) = 0, \quad \sum_{\gamma} \nu(\gamma) \gamma = 0, \quad \sum_{\gamma} \nu(\gamma) \gamma \gamma^{\top} = C. \quad (10)$$

Proof. The Laurent polynomial ring

$$R = \mathbb{Q}[X_1^{\pm 1}, \dots, X_k^{\pm 1}]$$

is the group algebra of Γ , with $X^{\gamma} = \prod_j X_j^{\gamma_j}$. The quotient map $\Gamma \rightarrow \Gamma/\Lambda_i$ induces a ring homomorphism whose kernel is

$$J_i = (X^h - 1 : h \in \Lambda_i).$$

Indeed, its kernel consists of Laurent polynomials with zero coefficient sum on each coset. Within a coset, choose a representative γ_0 from the finite support; a zero-sum polynomial on that coset is a sum of terms $c_{\gamma}(X^{\gamma} - X^{\gamma_0})$, each in J_i .

We will find a Laurent polynomial in every J_i with prescribed terms through degree two near $X = (1, \dots, 1)$. To use the tensor hypothesis, we pass to formal logarithmic coordinates, in which the generators $X^h - 1$ become linear forms multiplied by units. Put $\mathfrak{m} = (X_1 - 1, \dots, X_k - 1)$ and complete R with respect to $\mathfrak{m}$. With $Z_j = X_j - 1$, its completion is $\hat{R} = \mathbb{Q}[[Z_1, \dots, Z_k]]$: the factors $1 + Z_j$ are already invertible modulo each power of $(Z_1, \dots, Z_k)$. The formal changes of variables

$$Y_j = \log(1 + Z_j), \quad Z_j = \exp(Y_j) - 1$$

identify this ring with $\mathbb{Q}[[Y_1, \dots, Y_k]]$ and identify $\mathfrak{m}\hat{R}$ with $(Y_1, \dots, Y_k)$. For each $h \in \Gamma$,

$$X^h - 1 = \exp(h \cdot Y) - 1 = (h \cdot Y) \sum_{n \geq 0} \frac{(h \cdot Y)^n}{(n+1)!}.$$

The final factor is a unit. Hence

$$J_i \hat{R} = (h \cdot Y : h \in \Lambda_i) \hat{R}. \quad (11)$$

Consider the quadratic polynomial

$$Q_2(Y) = \frac{1}{2} Y^{\top} C Y.$$

Identify $\mathbb{Q}[Y_1, \dots, Y_k]$ with the symmetric algebra of $\mathbb{Q}^k$ by sending ε_j to Y_j . The map to the symmetric algebra of $\mathbb{Q}^k/V_i$ has kernel generated by V_i in degree one. This follows, for example, by extending a basis of V_i to a basis of $\mathbb{Q}^k$. The tensor hypothesis therefore puts Q_2 in the ideal generated by the linear forms $h \cdot Y$, $h \in \Lambda_i$. By (11), $Q_2 \in J_i \hat{R}$ for every i .

The ring R is Noetherian, being a localization of a polynomial ring over $\mathbb{Q}$. Its $\mathfrak{m}$ -adic completion is therefore flat over R [23, Lemma 10.97.2, Tag 00MB]. If $I = \bigcap_{i=1}^s J_i$, the injection

$$R/I \longrightarrow \bigoplus_{i=1}^s R/J_i$$

remains injective after tensoring with $\widehat{R}$. Consequently

$$I\widehat{R} = \bigcap_{i=1}^s J_i \widehat{R},$$

and $Q_2 \in I\widehat{R}$.

We now recover a finite Laurent polynomial while retaining exact membership in I . Write

$$Q_2 = \sum_{a=1}^N f_a \alpha_a, \quad f_a \in I, \quad \alpha_a \in \widehat{R},$$

as a finite sum. For each a , truncate the formal Z -expansion of α_a through degree two to obtain $b_a \in R$ with $b_a - \alpha_a \in \mathfrak{m}^3 \widehat{R}$. Then $f = \sum_a f_a b_a \in I$ and

$$f(e^{Y_1}, \dots, e^{Y_k}) = \frac{1}{2} Y^\top C Y \pmod{(Y_1, \dots, Y_k)^3}.$$

Write the finite Laurent expansion as $f = \sum_\gamma \nu(\gamma) X^\gamma$. Membership in each J_i gives (9). Finally,

$$X^\gamma = \exp(\gamma \cdot Y) = 1 + \gamma \cdot Y + \frac{1}{2}(\gamma \cdot Y)^2 \pmod{(Y)^3}.$$

Comparing the constant term, gradient, and Hessian at $Y = 0$ gives (10). $\square$

Apply Lemma 3.1 to the matrix and evaluation lattices constructed above. If $\gamma, \gamma' \in \Gamma$, then $e_i(w(\gamma)) = e_i(w(\gamma'))$ exactly when $\gamma - \gamma' \in \Lambda_i$. Thus (9) says that the positive and negative weights cancel separately at every evaluation value. The second moment in (10), together with (8), gives their prescribed difference in gradient Gram matrices. We next redistribute these weights so that the negative part has arbitrarily small gradient second moment, while all these equalities remain exact.

4 Exact coordinate permutations at two scales

The tensor condition will now produce two configurations with the same coordinate values at each label, but with arbitrarily different scales. The equality of coordinate values is an equality of finite multisets, including multiplicities.

Lemma 4.1 (Two-scale configurations). *Let $A = \{a_1, \dots, a_s\} \subset \mathbb{R}^d$ have affine span $\mathbb{R}^d$, with $s \geq 2$. Let $W = \mathbb{R}^{1+d}$ be regarded as a vector space over $\mathbb{Q}$. Suppose that a symmetric tensor $T \in W \otimes_{\mathbb{Q}} W$ satisfies*

$$(e_i \otimes e_i)(T) = 0 \quad (1 \leq i \leq s), \quad G(T) = I_d,$$

where $e_i(v, u) = v + a_i \cdot u$. For every $\eta > 0$, there are an integer $\ell \geq 1$ and vectors $f_i, g_i \in \mathbb{R}^\ell$ such that

$$\|f_i - f_j\|^2 = \|a_i - a_j\|^2, \quad \|g_i - g_j\|^2 = q \|a_i - a_j\|^2, \quad q = \frac{\eta}{1 + \eta}, \quad (12)$$

for every i, j , and g_i is a permutation of the coordinates of f_i for each i .

The finite function ν from Lemma 3.1 supplies signed rational weights whose sums vanish on every evaluation class. Positive and negative weights can therefore become two lists of coordinates with identical multisets of values. Their second moments control the difference of their gradient Gram matrices. The remaining task is to make the negative list's Gram matrix arbitrarily small. We first obtain the required inequality by smoothing, and then obtain finite rational weights while preserving all cancellation identities exactly.

For a real symmetric $k \times k$ matrix C , write

$$D_C = \frac{1}{2} \sum_{j,l=1}^k C_{jl} \partial_j \partial_l, \quad r_+ = \max\{r, 0\}, \quad r_- = \max\{-r, 0\} \quad (r \in \mathbb{R}).$$

Lemma 4.2 (A compactly supported smoothing function). *Let C be a real symmetric $k \times k$ matrix and let $U \in \text{Mat}_{d \times k}(\mathbb{R})$, where $d \geq 1$ and $UCU^\top = I_d$. For every $\eta > 0$, there is a nonnegative $b \in C_c^\infty(\mathbb{R}^k)$ such that*

$$\int_{\mathbb{R}^k} b(x) dx > 0, \quad \int_{\mathbb{R}^k} (D_C b)_-(x) \|Ux\|^2 dx < \eta \int_{\mathbb{R}^k} b(x) dx. \quad (13)$$

Proof. We will average Gaussian densities with covariance matrices running from S to $S + C$. We first construct a positive definite S for which $S + C$ is positive definite and $\text{tr}(USU^\top)$ is small.

The identity $UCU^\top = I_d$ implies that U has rank d . Set $H = \text{im}(U^\top)$ and $V = \ker U$, so that $\mathbb{R}^k = H \oplus V$ orthogonally. The restriction of the quadratic form C to H is positive definite: if $0 \neq h = U^\top y$, then

$$h^\top C h = y^\top U C U^\top y = \|y\|^2 > 0.$$

Choose $\epsilon > 0$ with $\epsilon \text{tr}(UU^\top) < \eta/8$. We claim that, for a sufficiently large $L > 0$, the matrix

$$S = \epsilon P_H + L P_V$$

is positive definite and so is $S + C$. Here P_H, P_V are the orthogonal projections. To check the latter assertion, write

$$S + C = \begin{pmatrix} A & B \\ B^\top & C_{VV} + L I_V \end{pmatrix}, \quad A = C_{HH} + \epsilon I_H > 0.$$

Choose L such that $C_{VV} + L I_V - B^\top A^{-1} B > 0$ and use the Schur complement. If $V = \{0\}$, the second block is absent. Since $U P_V = 0$, this construction also gives

$$\text{tr}(USU^\top) = \epsilon \text{tr}(UU^\top) < \eta/8. \quad (14)$$

Every matrix $S + tC$, $0 \leq t \leq 1$, is positive definite, being a convex combination of S and $S + C$.

For a positive definite $k \times k$ matrix Q , let

$$\phi_Q(x) = (2\pi)^{-k/2} (\det Q)^{-1/2} \exp\left(-\frac{1}{2} x^\top Q^{-1} x\right)$$

be the centered Gaussian density with covariance Q . Differentiating the displayed formula, with $Q_t = S + tC$, gives

$$\frac{d}{dt} \phi_{Q_t}(x) = \frac{1}{2} \left(x^\top Q_t^{-1} C Q_t^{-1} x - \text{tr}(Q_t^{-1} C) \right) \phi_{Q_t}(x) = D_C \phi_{Q_t}(x).$$

This identity does not require C to be positive semidefinite. The covariances Q_t have uniformly bounded eigenvalues, bounded away from zero, so the densities and each of their spatial derivatives have uniform Gaussian decay. Consequently

$$b_*(x) = \int_0^1 \phi_{S+tC}(x) dt$$

is a nonnegative Schwartz function of integral 1, and differentiation under the integral gives

$$D_C b_* = \phi_{S+C} - \phi_S.$$

The pointwise inequality $(\phi_{S+C} - \phi_S)_- \leq \phi_S$ and (14) imply

$$\int_{\mathbb{R}^k} (D_C b_*)_-(x) \|Ux\|^2 dx \leq \int_{\mathbb{R}^k} \phi_S(x) \|Ux\|^2 dx = \text{tr}(USU^\top) < \eta/8. \quad (15)$$

To make the support compact, choose a smooth function χ with $0 \leq \chi \leq 1$, equal to 1 on the unit ball and zero outside the ball of radius 2. Put $\chi_R(x) = \chi(x/R)$ and $b_R = \chi_R b_*$. Then $\int b_R \rightarrow 1$. The product rule reads

$$D_C b_R = \chi_R D_C b_* + \sum_{j,l} C_{jl} (\partial_j \chi_R) (\partial_l b_*) + b_* D_C \chi_R.$$

The derivatives of χ_R have bounds of orders R^{-1} and R^{-2} and are supported where $R \leq \|x\| \leq 2R$. The Schwartz decay therefore implies

$$\int_{\mathbb{R}^k} |D_C b_R - D_C b_*(x)| \|Ux\|^2 dx \rightarrow 0.$$

Since $r \mapsto r_-$ is 1-Lipschitz, the corresponding weighted negative-part integrals converge as well. For sufficiently large R , we have $\int b_R > 1/2$ and the last displayed integral is less than $\eta/8$. Equation (15) now gives

$$\int (D_C b_R)_- \|Ux\|^2 < \eta/4 < \eta \int b_R.$$

Taking $b = b_R$ proves the lemma. $\square$

Lemma 4.3 (Finite rational weights). *Let $\Gamma = \mathbb{Z}^k$, and let $\Lambda_1, \dots, \Lambda_s$ be subgroups of Γ . Suppose that a finitely supported function $\nu : \Gamma \rightarrow \mathbb{Q}$ has zero pushforward on each Γ/Λ_i , and satisfies*

$$\sum_{\gamma} \nu(\gamma) = 0, \quad \sum_{\gamma} \nu(\gamma) \gamma = 0, \quad \sum_{\gamma} \nu(\gamma) \gamma \gamma^\top = C.$$

Let U be a real $d \times k$ matrix with $UCU^\top = I_d$, where $d \geq 1$. For every $\eta > 0$, there are an integer $n \geq 1$, a finitely supported $\mu : \Gamma \rightarrow \mathbb{Q}$, and $\zeta \in \mathbb{Q}_{>0}$ such that μ has zero pushforward on each Γ/Λ_i and

$$\sum_{\lambda} \mu(\lambda) \frac{\lambda}{n} \frac{\lambda^\top}{n} = \zeta C, \quad (16)$$

$$\sum_{\lambda} \mu(\lambda)_- \left\| U \frac{\lambda}{n} \right\|^2 < \eta \zeta. \quad (17)$$

Proof. Take b from Lemma 4.2. For each positive integer n , set

$$d_n(\lambda) = n^{2-k} b(\lambda/n), \quad \mu_n(\lambda) = (\nu * d_n)(\lambda) = \sum_{\gamma} \nu(\gamma) d_n(\lambda - \gamma).$$

These functions have finite support. Quotient pushforward commutes with convolution, so every required pushforward of μ_n is zero. More generally, for any finitely supported real function d on Γ , the moment identities for ν give

$$\sum_{\lambda} (\nu * d)(\lambda) = 0, \quad \sum_{\lambda} (\nu * d)(\lambda) \lambda \lambda^\top = \left(\sum_{\beta} d(\beta) \right) C. \quad (18)$$

Indeed, substitute $\lambda = \beta + \gamma$ in the double sum. The terms in $\beta\beta^\top$, $\beta\gamma^\top$ and $\gamma\beta^\top$ vanish by the zeroth and first moments of ν . Thus, with

$$\zeta_n = n^{-2} \sum_{\lambda} d_n(\lambda) = n^{-k} \sum_{\lambda} b(\lambda/n),$$

the identity in (16) holds exactly for (μ_n, ζ_n) , and Riemann sums give $\zeta_n \rightarrow \int b > 0$.

For $x = \lambda/n$, Taylor expansion gives

$$\begin{aligned} n^k \mu_n(\lambda) &= n^2 \sum_{\gamma} \nu(\gamma) b(x - \gamma/n) \\ &= \frac{1}{2} \sum_{j,l} C_{jl} \partial_j \partial_l b(x) + O(n^{-1}) = D_C b(x) + O(n^{-1}). \end{aligned}$$

The error is uniform in x : the third derivatives of b are bounded, and $\sum_{\gamma} |\nu(\gamma)| \|\gamma\|^3$ is finite. Moreover, the supports of both $\mu_n(\lambda)$ and $D_C b(\lambda/n)$, after division of the lattice by n , lie in a single fixed ball for all $n \geq 1$. There are $O(n^k)$ relevant lattice points and the weight $\|Ux\|^2$ is bounded on that ball. The 1-Lipschitz property of the negative part therefore shows that

$$\sum_{\lambda} \mu_n(\lambda)_{-} \|U\lambda/n\|^2 - n^{-k} \sum_{\lambda} (D_C b(\lambda/n))_{-} \|U\lambda/n\|^2 = O(n^{-1}).$$

The second sum is a Riemann sum for a continuous compactly supported function. By (13), it follows that

$$\sum_{\lambda} \mu_n(\lambda)_{-} \|U\lambda/n\|^2 \longrightarrow \int (D_C b)_{-} \|Ux\|^2 < \eta \int b.$$

Fix a sufficiently large n such that $\zeta_n > 0$ and (17) holds strictly for (μ_n, ζ_n) .

It remains to obtain rational coefficients. Let $E \subset \Gamma$ be a finite set supporting d_n . For functions d supported in E , the maps

$$d \longmapsto \nu * d, \quad d \longmapsto \zeta = n^{-2} \sum_{\lambda} d(\lambda)$$

are linear, and the function

$$d \longmapsto \sum_{\lambda \in \text{supp } \nu + E} (\nu * d)(\lambda)_{-} \|U\lambda/n\|^2$$

is continuous on $\mathbb{R}^E$. The two strict inequalities $\zeta > 0$ and (17) consequently persist in an open neighborhood of d_n . Choose a rational vector d in that neighborhood and extend it by zero outside E . Put $\mu = \nu * d$. Both μ and ζ are rational. Convolution still gives zero quotient pushforwards, and (18) still gives the exact second moment (16). This proves the lemma. $\square$

We have now obtained finite rational weights with exact cancellation and the required second moment. The inequalities will only be used to add a common positive semidefinite Gram increment; no further approximation is needed.

Proof of Lemma 4.1. Use the notation preceding Lemma 3.1 and the function ν supplied by that lemma: $w_j = (v_j, u_j)$, $w(\lambda) = \sum_j \lambda_j w_j$, $U = (u_1 \cdots u_k)$, $\Gamma = \mathbb{Z}^k$, and

$$\Lambda_i = \{\lambda \in \Gamma : e_i(w(\lambda)) = 0\}.$$

The rational symmetric matrix C of the tensor satisfies $UCU^\top = I_d$. Apply Lemma 4.3 to obtain n, μ, ζ . Choose a positive integer M such that every $M\mu(\lambda)$ is an integer. Form two

finite lists of affine rows: the positive list has $M\mu(\lambda)_+$ repetitions of $w(\lambda/n)$, and the negative list has $M\mu(\lambda)_-$ repetitions.

The zero total mass of μ shows that the two lists have the same length. For each label i ,

$$e_i(w(\lambda/n)) = e_i(w(\lambda'/n)) \iff \lambda - \lambda' \in \Lambda_i.$$

Thus the zero pushforward on Γ/Λ_i says exactly that, after evaluating at a_i , every real coordinate value appears equally often in the two lists. The resulting evaluation vectors are coordinate permutations of one another.

Divide every affine row in both lists by $\sqrt{\zeta M}$. Their gradient Gram matrices become

$$B_{\pm} = \frac{1}{\zeta} \sum_{\lambda} \mu(\lambda)_{\pm} (U\lambda/n)(U\lambda/n)^{\top}.$$

Equations (16) and (17) give

$$B_+ - B_- = I_d, \quad \text{tr}(B_-) < \eta.$$

Since B_- is positive semidefinite, each of its eigenvalues is at most its trace. Hence $P = \eta I_d - B_-$ is positive semidefinite. Choose vectors $r_1, \dots, r_d \in \mathbb{R}^d$ with $\sum_j r_j r_j^{\top} = P$, for example the columns of $P^{1/2}$, and append the same affine rows $(0, r_j)$ to both lists. The two Gram matrices are now $(1 + \eta)I_d$ and ηI_d . Identical appended rows preserve equality of all coordinate multisets.

Finally divide every row in both lists by $\sqrt{1 + \eta}$, and let f_i, g_i be their respective evaluation vectors at a_i . Their coordinate multisets remain equal. For any affine row (v, u) , the difference of its evaluations at a_i and a_j is $(a_i - a_j) \cdot u$. The two gradient Gram identities therefore give (12) exactly, with $q = \eta/(1 + \eta)$. $\square$

5 From paths to monochromatic copies

We now give a criterion that converts products of scaled copies into the Ramsey property. The products will first be taken in a free group. A word construction then turns their algebraic relations into actual monochromatic point sets. Throughout this section, $A = \{a_1, \dots, a_s\}$ is a finite Euclidean configuration with $s \geq 2$.

Let $\mathcal{E} = \mathbb{R}^{(\mathbb{N})}$ be the Euclidean space of real sequences with finite support, and let K be the free group on the symbols ξ_z , indexed by $z \in \mathcal{E}$. Write $\Delta g = (g, \dots, g) \in K^s$ for $g \in K$. A *path* is a chosen finite sequence of factors in K^s of the following two kinds:

1. a diagonal factor Δg , assigned weight zero;
2. a factor $X_b = (\xi_{b_1}, \dots, \xi_{b_s})$, where $b_1, \dots, b_s \in \mathcal{E}$ and, for some $u \geq 0$,

$$\|b_i - b_j\|^2 = u \|a_i - a_j\|^2 \quad (1 \leq i, j \leq s),$$

assigned weight u .

The *endpoint* is the product of the factors in K^s , in their given order. The *weight* is the sum of their assigned weights. The weight belongs to the chosen path, not to its endpoint. In particular, two paths with the same endpoint can have different weights.

Proposition 5.1 (A path criterion). *Suppose that for every integer $n \geq 1$ there are two paths with the same endpoint and weights $\lambda > 0$ and $\mu \geq 0$ satisfying $\mu/\lambda < 1/n$. Then A is Euclidean Ramsey.*

We prove the proposition in three steps. We first reduce to colorings of $\mathcal{E}$ and synchronize the available path weights. We then develop the ultrafilter facts needed to realize group products by finite strings. Finally, the Hales–Jewett theorem chooses the variable positions of a monochromatic string family.

5.1 Finite dimensions and synchronized weights

The following finite-dimensional reduction is the compactness principle used in Euclidean Ramsey theory; compare [4, Proposition 4]. We include its proof for arbitrary colorings.

Lemma 5.2. *Fix an integer $r \geq 2$. If every r -coloring of $\mathcal{E}$ contains a monochromatic congruent copy of A , then the same holds for every r -coloring of $\mathbb{R}^D$, for some finite D .*

Proof. Suppose instead that every $\mathbb{R}^D$ admits an r -coloring avoiding A . Consider the compact product space $\{1, \dots, r\}^{\mathcal{E}}$. For each ordered congruent copy $(b_1, \dots, b_s)$ of A in $\mathcal{E}$, impose the condition that the colors of the b_i are not all equal. This is a clopen condition depending on finitely many coordinates of the product space.

Every finite collection of these conditions involves finitely many finite-support vectors. They all lie in the first D coordinate positions for some D . An avoiding coloring of this $\mathbb{R}^D$, extended arbitrarily to $\mathcal{E}$, satisfies the finite collection. Compactness therefore gives a coloring satisfying all the conditions, contrary to the hypothesis. $\square$

Lemma 5.3. *Fix an integer $n \geq 1$. Let two paths have the same endpoint and weights $\lambda > 0$ and $\mu \geq 0$. If $\rho = \mu/\lambda < 1/n$, then there is an endpoint $w \in K^s$ admitting a path of weight $1/t$ for every integer $1 \leq t \leq n$.*

Proof. Set

$$\alpha_t = \frac{1/t - \rho}{1 - \rho} \quad (1 \leq t \leq n).$$

These numbers satisfy $0 < \alpha_n < \dots < \alpha_1 = 1$. Partition $[0, 1]$ at these points. For each resulting interval of length $l > 0$, apply to both paths the endomorphism of K defined by

$$\xi_z \mapsto \xi_{\sqrt{l/\lambda}z}.$$

It preserves diagonal factors and multiplies every copy-factor weight by l/λ . The two new paths thus have the same endpoint and weights l and ρl .

Concatenate the interval paths in increasing interval order. The resulting endpoint is independent of which of the two paths is chosen on each interval. For a given t , choose the larger weight on the intervals before α_t and the smaller weight on the remaining intervals. The total weight is

$$\alpha_t + \rho(1 - \alpha_t) = \frac{1}{t}.$$

The fixed order of concatenation gives one endpoint w for all t ; no commutation of factors is used. $\square$

5.2 Realizing ultrafilter products by strings

Let $\mathcal{S}$ be the monoid of finite strings on the alphabet $\mathcal{E}$, with concatenation as multiplication and the empty string ϵ as identity. Choose pairwise disjoint infinite coordinate blocks in $\mathcal{E}$, and let $J_j : \mathcal{E} \rightarrow \mathcal{E}$ be the corresponding linear isometric embeddings, one for each $j \geq 1$. For a finite string $z = (z_1, \dots, z_L)$, define

$$\Phi(z) = \sum_{j=1}^L J_j z_j.$$

For two strings of the same length,

$$\|\Phi(z) - \Phi(z')\|^2 = \sum_{j=1}^L \|z_j - z'_j\|^2. \quad (19)$$

Thus we seek a realization in which each copy factor supplies its prescribed one-letter strings, while each diagonal factor supplies one common string in all s roles. Common strings will contribute zero to the squared distances; the remaining contribution will be the path weight times the corresponding squared distance in A . Diagonal factors may contain group inverses, so they cannot in general be evaluated directly in the string monoid. We instead map K into a group of ultrafilters on $\mathcal{S}$ and then realize its common factors by common strings.

The following standard construction fixes the multiplication convention; see [9, Theorem 2.5, Corollary 2.6, and Theorem 1.42]. The classical idempotent argument also appears in Ellis's semigroup lemma [3, Lemma 1]; taking the opposite semigroup matches the one-sided continuity convention used here.

An ultrafilter on $\mathcal{S}$ is a maximal proper filter of subsets of $\mathcal{S}$. Write $\beta\mathcal{S}$ for the space of all such ultrafilters. For $B \subseteq \mathcal{S}$, its basic clopen set is

$$\overline{B} = \{u \in \beta\mathcal{S} : B \in u\}.$$

This space is compact and Hausdorff: the ultrafilter conditions define a closed subset of $\{0, 1\}^{\mathcal{P}(\mathcal{S})}$. For a string x , put $x^{-1}B = \{y \in \mathcal{S} : xy \in B\}$; this notation denotes a preimage under concatenation, not a string inverse. Define the product of ultrafilters by

$$B \in uv \iff \{x \in \mathcal{S} : x^{-1}B \in v\} \in u. \quad (20)$$

The principal ultrafilter at x is denoted by δ_x .

Lemma 5.4. *The operation in (20) makes $\beta\mathcal{S}$ a compact right-topological semigroup with identity δ_ϵ . It contains an idempotent p such that $p(\beta\mathcal{S})p$ is a group with identity p .*

Proof. For fixed v , the map $B \mapsto \{x : x^{-1}B \in v\}$ preserves finite intersections and complements. Thus (20) defines an ultrafilter uv . For $u, v, w \in \beta\mathcal{S}$, both parenthesizations of their product contain B precisely when

$$\{x : \{y : \{z : xyz \in B\} \in w\} \in v\} \in u.$$

This proves associativity. For fixed v , the inverse image of $\overline{B}$ under $u \mapsto uv$ is $\overline{\{x : x^{-1}B \in v\}}$, so right translation is continuous. The assertion about δ_ϵ follows directly from the definition.

Choose a minimal nonempty compact left ideal L_0 in $\beta\mathcal{S}$. Such an ideal exists because every chain of nonempty compact left ideals has a nonempty intersection. For each $v \in L_0$, the set $(\beta\mathcal{S})v$ is a nonempty compact left ideal contained in L_0 , hence

$$(\beta\mathcal{S})v = L_0. \quad (21)$$

The ideal L_0 is also a compact subsemigroup.

Choose a minimal nonempty compact subsemigroup $H_0 \subseteq L_0$. For $x \in H_0$, the set H_0x is compact and is a subsemigroup: $(ax)(bx) = a(xb)x \in H_0x$ for $a, b \in H_0$. Minimality gives $H_0x = H_0$. Consequently

$$\{y \in H_0 : yx = x\}$$

is nonempty, compact, and a subsemigroup. It equals H_0 by minimality, so $x^2 = x$. In particular L_0 contains an idempotent p .

The corner $G = p(\beta\mathcal{S})p$ is a monoid with identity p . Every $y \in G$ lies in L_0 , so (21) gives $p = vy$ for some $v \in \beta\mathcal{S}$. Since $py = y$,

$$(pvp)y = pvy = p.$$

Thus every element of G has a left inverse in G . A monoid with this property is a group: if $ay = p$ and $ba = p$, then $b = b(ay) = (ba)y = y$, whence $ya = p$ as well. $\square$

The next lemma explains why common ultrafilter factors can be replaced by common strings. This is the step that will preserve the distances between the different roles in a monochromatic configuration.

Lemma 5.5 (Simultaneous realization). *Let $q_{i,j} \in \beta\mathcal{S}$ for $1 \leq i \leq s$ and $1 \leq j \leq M$, where $M \geq 1$. Suppose that every column j has one of the following specified forms:*

1. $q_{i,j} = q_j$ is independent of i ;
2. $q_{i,j} = \delta_{z_{i,j}}$ for prescribed strings $z_{i,j} \in \mathcal{S}$.

If $B \in q_{i,1} \cdots q_{i,M}$ for every i , there are strings $x_{i,j}$ such that all products $x_{i,1} \cdots x_{i,M}$ belong to B , with $x_{i,j}$ independent of i in each column of the first form and $x_{i,j} = z_{i,j}$ in each column of the second form.

Proof. Choose the strings from left to right. Before column j , let P_i be the already chosen prefix and maintain the invariant

$$P_i^{-1}B \in q_{i,j} \cdots q_{i,M}.$$

Initially $P_i = \epsilon$. Put $v_i = q_{i,j+1} \cdots q_{i,M}$, interpreting an empty product here as δ_ϵ . By (20), the set

$$E_i = \{x : x^{-1}(P_i^{-1}B) \in v_i\}$$

belongs to $q_{i,j}$. In a common column, the finite intersection $\bigcap_i E_i$ belongs to the common ultrafilter and is nonempty; choose the same string from it for all i . In a prescribed principal column, $z_{i,j} \in E_i$, so choose that string. Replacing P_i by $P_i x_{i,j}$ preserves the invariant. After the last column it says $P_i^{-1}B \in \delta_\epsilon$, which is precisely $P_i \in B$. $\square$

We now have both ingredients for the word construction: a group in which the path endpoints can be evaluated, and a way to realize their products by strings while choosing every common factor equally in all roles.

5.3 The Hales–Jewett step

Proof of Proposition 5.1. Fix $r \geq 2$ and an arbitrary coloring $c : \mathcal{E} \rightarrow \{1, \dots, r\}$. Choose n such that every r -coloring of $\{1, \dots, s\}^n$ has a monochromatic combinatorial line, by the Hales–Jewett theorem [8]; see also Shelah’s proof and primitive-recursive bound [22, p. 686 and Section 1]. We use only existence of the length. Such a line has a nonempty set of variable positions, occupied by one common letter, and fixed letters in all other positions.

In the group formulation of Leader, Russell and Walters [16, Conjecture C and the following discussion], the number of varying positions is prescribed before coloring; without that restriction, ordinary Hales–Jewett applies. Here we allow the line to choose that number and prepare all the required path weights in advance. The hypothesis and Lemma 5.3 give an endpoint $w = (w_1, \dots, w_s) \in K^s$ and a path of weight $1/t$ ending at w for every $1 \leq t \leq n$.

Color $\mathcal{S}$ by $c \circ \Phi$, and write its color classes as $B_1, \dots, B_r$.

Choose p as in Lemma 5.4. The freeness of K gives a homomorphism

$$\psi : K \longrightarrow p(\beta\mathcal{S})p, \quad \psi(\xi_z) = p\delta_{(z)}p,$$

where (z) is the one-letter string with entry z . Color a word $i_1 \cdots i_n$ by the unique index k for which

$$B_k \in \psi(w_{i_1}) \cdots \psi(w_{i_n}).$$

Every ultrafilter contains exactly one class of a finite partition, so this defines an r -coloring.

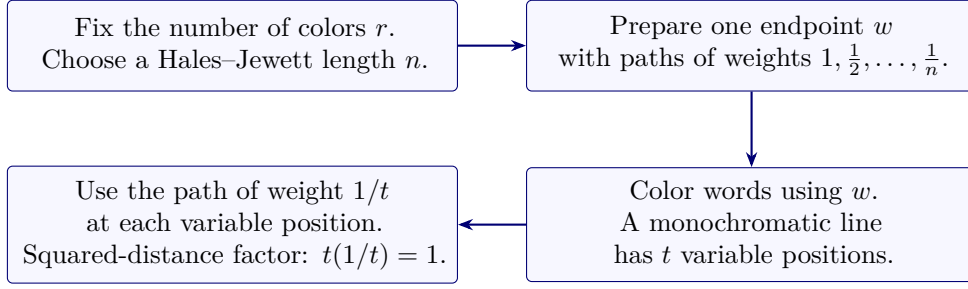


Figure 1: The order of choices in the path criterion. Every possible variable count is accommodated before the word coloring is defined. Choosing a path representation later leaves its endpoint unchanged; common factors become common strings, so only the variable blocks contribute to pairwise distances.

Take a monochromatic combinatorial line and let t be its number of variable positions. Figure 1 records the order of these choices.

For each of the line’s s words, expand the factor at every variable position using the chosen path ending at w with weight $1/t$. A diagonal path factor contributes one common ultrafilter $\psi(g)$. A copy factor contributes $p\delta_{(b_i)}p$ in role i : the two p factors are common and the middle factor is principal at a prescribed one-letter string. Unexpanded fixed positions are common factors as well. The identity of the corner group is p ; if an empty group product occurs, it is retained as this common factor. The expanded products therefore have the column forms required by Lemma 5.5, and all contain the same color class B_k .

Apply that lemma. It produces strings $z^{(1)}, \dots, z^{(s)}$ in B_k . Every common factor was realized by the same string in all roles, and every varying factor has length one. Hence the resulting strings have equal lengths and their entries align. Common strings contribute zero to their pairwise squared distances. Each of the t expanded paths contributes $\|a_i - a_j\|^2/t$ by its weight. Equation (19) consequently gives

$$\|\Phi(z^{(i)}) - \Phi(z^{(j)})\|^2 = t \frac{1}{t} \|a_i - a_j\|^2 = \|a_i - a_j\|^2.$$

Their images form a monochromatic congruent copy of A in $\mathcal{E}$. Lemma 5.2 now supplies a finite forcing dimension for this r . Since r was arbitrary, A is Euclidean Ramsey. $\square$

6 From coordinate permutations to equal path endpoints

We now turn the two images of Lemma 4.1 into the equal-endpoint paths required by Proposition 5.1. Recall that $\mathcal{E} = \mathbb{R}^{(\mathbb{N})}$, that K is the free group on the letters ξ_z for $z \in \mathcal{E}$, and that path weights count the squared-distance scales of their copy factors. The two images give paths of very different weights. Their endpoints need not agree. Our task is to correct those endpoints at a cost small relative to the larger weight.

6.1 Corrections supported on individual coordinates

Let $M \subset K^s$ be the monoid generated by the diagonal subgroup ΔK and the tuples $X_b = (\xi_{b_i})_{i=1}^s$ associated to ordered congruent copies b of A . Thus a chosen expression for an element of M is a path using only copy factors of weight one. Its *degree* is the number of those factors; diagonal factors have degree zero. Degrees always refer to chosen expressions.

Suppose two paths have endpoints $D_+, D_- \in K^s$. If we can write

$$D_+^{-1}D_- = uv^{-1} \quad (u, v \in M),$$

then $D_+u = D_-v$. Appending the chosen expressions for u and v therefore gives equal endpoints, with added weights $\deg u$ and $\deg v$, respectively. We will make their sum small relative to the larger original path weight, so that correcting the endpoints retains the small ratio of weights.

To construct these corrections, let

$$N = \ker(K \longrightarrow \mathbb{Z}, \xi_z \longmapsto 1).$$

For distinct labels i, j , write M_{ij} for the image of M under projection onto coordinates i, j .

Lemma 6.1. *For every $i \neq j$, the monoid M_{ij} is a group containing $N \times N$.*

Proof. Put $r = \|a_i - a_j\| > 0$. Every ordered pair $z, z' \in \mathcal{E}$ at distance r extends to an ordered congruent copy of A . Indeed, send the direction of $a_j - a_i$ to $(z' - z)/r$, extend it to an orthonormal frame for the affine span of A , and translate to send a_i to z . The frame can be chosen inside a finite-dimensional coordinate subspace of $\mathcal{E}$.

Set $x = \xi_z$ and $y = \xi_{z'}$. Both (x, y) and (y, x) therefore belong to M_{ij} , and

$$(x, y)^{-1} = (x^{-1}, x^{-1})(y, x)(y^{-1}, y^{-1}) \in M_{ij}.$$

The diagonal generators are also invertible within M_{ij} , so this monoid is a group. It contains

$$(x, y)(y^{-1}, y^{-1}) = (xy^{-1}, 1), \quad (x, y)(x^{-1}, x^{-1}) = (1, yx^{-1}),$$

and their conjugates by diagonal elements.

The graph on $\mathcal{E}$ joining points at distance r is connected. To see this, subdivide any displacement into pieces of length at most $2r$. The endpoints of each piece have a common point at distance r : choose an appropriate perpendicular displacement from their midpoint. An additional coordinate supplies the perpendicular direction if needed. Differences of letters along graph edges consequently generate, after taking their normal closure in K , all differences $\xi_z \xi_{z'}^{-1}$. That normal closure is N , since identifying all free generators gives the infinite cyclic group. The two displayed types of elements and diagonal conjugation thus give $N \times N \subset M_{ij}$. $\square$

For $g \in K$, let $g^{[i]} \in K^s$ have value g in coordinate i and identity in every other coordinate. Define

$$L_i = \{g \in K : g^{[i]} = uv^{-1} \text{ for some } u, v \in M\}.$$

We call uv^{-1} a *right fraction*. Its cost, with chosen expressions for u, v , is the sum of their degrees. For $g \in L_i$, let $c_i(g)$ be the minimum cost of a right fraction representing $g^{[i]}$. The minimum exists because its possible costs form a nonempty set of nonnegative integers.

We will only multiply right fractions when their supports permit an explicit construction of a new denominator. No closure assertion for the whole set MM^{-1} is needed.

Lemma 6.2. *Each L_i is a normal subgroup of K , and its cost satisfies*

$$c_i(g^{-1}) = c_i(g), \quad c_i(kgk^{-1}) = c_i(g), \quad c_i(gh) \leq c_i(g) + c_i(h)$$

for $g, h \in L_i$ and $k \in K$. Every tuple $(g_i)_{i=1}^s$ with $g_i \in L_i$ has a represented right fraction of cost at most $\sum_i c_i(g_i)$. Moreover,

$$N^{(s-2)} \subset L_i \quad (1 \leq i \leq s), \tag{22}$$

where $N^{(0)} = N$ and $N^{(r+1)} = [N^{(r)}, N^{(r)}]$.

Proof. Inversion of a right fraction interchanges its numerator and denominator. If $d = \Delta(k)$, then

$$(kgk^{-1})^{[i]} = (du)(dv)^{-1} \quad \text{whenever} \quad g^{[i]} = uv^{-1}.$$

These operations preserve the represented cost. Applying each operation in both directions proves inversion and conjugation invariance.

The following multiplication rule supplies the cost estimate. Given any represented right fraction uv^{-1} and $g \in L_i$, put $h = v_i^{-1}gv_i$ and choose $h^{[i]} = ab^{-1}$ of cost $c_i(h) = c_i(g)$. Then

$$uv^{-1}g^{[i]} = uh^{[i]}v^{-1} = uab^{-1}v^{-1} = (ua)(vb)^{-1}. \quad (23)$$

The resulting cost is at most the previous cost plus $c_i(g)$. This proves product closure and subadditivity for L_i ; together with inversion and conjugation it proves normality. Successively applying (23) to the single-coordinate factors of $(g_i)_i$ proves the asserted tuple cost bound.

For the derived subgroup inclusion, we also need a multiplication rule for fractions supported on a pair of coordinates. Suppose that $z = cd^{-1}$ is such a fraction, supported on a pair J , and let uv^{-1} be any right fraction. Since the projection of M onto J is a group, choose $b \in M$ such that $t = vb$ is the identity on J . The tuples t and z commute, because their supports are disjoint. Consequently

$$uv^{-1}z = ubt^{-1}z = ubzt^{-1} = ubcd^{-1}t^{-1} = (ubc)(td)^{-1}. \quad (24)$$

This is a right fraction of finite cost. We require no bound on the cost of the auxiliary b .

We now prove (22) by induction on the number of coordinates. The proof applies to any submonoid of K^s containing the diagonal whose pair projections are groups containing $N \times N$. For $s = 2$, the whole monoid is its pair projection and contains $N \times N$, which proves the claim. For $s \geq 3$, fix distinct indices i, j, k . The monoid obtained by projecting away j satisfies the same abstract hypotheses. By induction, every $a \in N^{(s-3)}$ has a right fraction in that projection supported at i . Lift its numerator and denominator to M . Their quotient is a right fraction z_a supported on $\{i, j\}$, with value a at i . Projecting away k similarly gives a right fraction z_b supported on $\{i, k\}$ with any prescribed $b \in N^{(s-3)}$ at i .

The four factors of $z_a z_b z_a^{-1} z_b^{-1}$ are pair-supported right fractions; inverses remain fractions by interchanging numerator and denominator. Repeated use of (24) makes their product a right fraction. Its only possibly nonidentity coordinate is i , where its value is $[a, b]$. Thus $[a, b] \in L_i$. Since L_i is a subgroup, it contains the subgroup generated by all these commutators, namely $N^{(s-2)}$. $\square$

Set

$$L = N^{(s-2)}, \quad c(g) = \max_{1 \leq i \leq s} c_i(g) \quad (g \in L). \quad (25)$$

The subgroup L is normal in K . Lemma 6.2 shows that c is finite, symmetric under inversion, subadditive, and invariant under conjugation by every element of K . We have therefore identified a subgroup of allowable endpoint corrections and a cost for each one. We next construct long paths for which the needed corrections have small cost relative to their degree.

6.2 Averaging over coordinate permutations

Let H be a finite group of coordinate permutations acting on $\mathcal{E}$, and let $Z \subset \mathcal{E}$ be a finite H -invariant set. Consider represented tuples $w = (w_x)_{x \in Z} \in K^Z$ obtained by multiplying diagonal factors and factors

$$(\xi_{hx})_{x \in Z}, \quad h \in H.$$

The degree of such an expression is the number of factors of the latter kind. Right multiplication by a diagonal element has degree zero. Replacing w_x by w_{hx} preserves the degree: it sends a generator indexed by $g \in H$ to the generator indexed by gh .

Lemma 6.3. *Let H be a finite group of coordinate permutations of $\mathcal{E}$, and let $Z \subset \mathcal{E}$ be a finite H -invariant set. For every $\delta > 0$, there is a tuple $w \in K^Z$ represented by diagonal factors and factors $(\xi_{hx})_{x \in Z}$, $h \in H$, with degree $m > 0$, for which*

$$w_x w_y^{-1} \in L, \quad c(w_x w_y^{-1}) \leq \delta m \quad \text{whenever } x, y \text{ belong to the same } H\text{-orbit.} \quad (26)$$

Proof. We first arrange membership in L , then make the costs small. Start with $w_x = \xi_x$, represented with degree one; its differences $w_x w_y^{-1}$ belong to N .

Suppose the differences on an orbit O belong to $N^{(r)}$. Choose $x_0 \in O$ and right-multiply by the diagonal element $w_{x_0}^{-1}$. After this operation, every value on O belongs to $N^{(r)}$, and every difference on every orbit is unchanged. Replace the resulting tuple by

$$w'_x = \prod_{h \in H} w_{hx}, \quad (27)$$

using a fixed order on H . For $x \in O$, the list $(hx)_{h \in H}$ visits every point of O with the same multiplicity $|H|/|O|$. The products in (27) therefore have the same image in the abelian group $N^{(r)}/N^{(r+1)}$ for every $x \in O$. Their differences belong to $N^{(r+1)}$.

This procedure preserves any normal-subgroup condition already obtained on another orbit. Indeed, if $w_x w_y^{-1} \in G$ there for a normal subgroup $G \triangleleft K$, corresponding factors of the two products have the same image in K/G . Their products have the same image as well. Process the finitely many orbits, increasing the derived depth on each to $s - 2$. Every degree stays positive, and we obtain membership in L for all within-orbit differences. When $s = 2$, the initial tuple already has this property.

Fix one orbit O . Right-normalize at $x_0 \in O$ once more, so all values on O lie in L , and write $B = \max_{x \in O} c(w_x)$. Put $h = |H|$. If $h = 1$, all orbits are singletons and (26) already holds. Hence suppose $h \geq 2$. Repeatedly perform the operation

$$w'_x = \left(\prod_{g \in H} w_{gx} \right) w_{x_0}^{-1}. \quad (28)$$

For each $x \in O$, one factor in the product is $a = w_{x_0}$. Write the product as PaQ . Then

$$w'_x = P(aQa^{-1}), \quad c(w'_x) \leq c(P) + c(Q) \leq (h - 1)B.$$

All new values on O still lie in L , and their represented degree is h times the preceding degree. Thus, after k repetitions starting from degree m_0 , the maximum difference cost divided by degree is at most

$$\frac{2B}{m_0} \left(\frac{h - 1}{h} \right)^k. \quad (29)$$

Here the factor two uses $c(w_x w_y^{-1}) \leq c(w_x) + c(w_y)$. The bound tends to zero.

It remains to check that treating this orbit does not spoil a bound on another orbit. If $a_j b_j^{-1} \in L$ for $1 \leq j \leq h$, conjugation invariance and subadditivity give

$$c((a_1 \cdots a_h)(b_1 \cdots b_h)^{-1}) \leq \sum_{j=1}^h c(a_j b_j^{-1}). \quad (30)$$

For completeness, write $A_j = a_1 \cdots a_j$ and $B_j = b_1 \cdots b_j$, with $A_0 = B_0 = 1$. The identity

$$A_j B_j^{-1} = (A_{j-1} (a_j b_j^{-1}) A_{j-1}^{-1}) (A_{j-1} B_{j-1}^{-1})$$

proves (30) inductively. Consequently the product in (28) multiplies the maximum difference cost on any other orbit by at most h , while multiplying the degree by exactly h . The appended common right factor does not change differences. Their cost divided by degree cannot increase. We may therefore achieve the bound δ on one orbit after another; there are only finitely many orbits. $\square$

6.3 Equal endpoints and completion of sufficiency

We can now compare the two paths supplied by coordinate permutations. The preceding lemma makes their endpoint discrepancies inexpensive, and Lemma 6.2 realizes all discrepancies by actual paths of unit-scale copies.

Proposition 6.4. *Suppose $q > 0$ and $f_i, g_i \in \mathbb{R}^\ell$ satisfy*

$$\|f_i - f_j\|^2 = \|a_i - a_j\|^2, \quad \|g_i - g_j\|^2 = q\|a_i - a_j\|^2 \quad (1 \leq i, j \leq s),$$

and each g_i is a coordinate permutation of f_i . For every $\delta > 0$ there exist two paths with the same endpoint and weights $\lambda > 0$ and $\mu \geq 0$ such that

$$\frac{\mu}{\lambda} \leq q + s\delta.$$

Proof. Embed $\mathbb{R}^\ell$ into $\mathcal{E}$, let H be its full coordinate permutation group, acting trivially on the remaining coordinates, and let Z be the union of the H -orbits of the f_i and g_i . It is finite, and f_i, g_i lie in the same orbit for each i . Apply Lemma 6.3 with the given δ and let $m > 0$ be the degree of its represented tuple w .

Restricting the same expression to the labels f_i gives the endpoint $D_+ = (w_{f_i})_{i=1}^s$ of a path of weight m . Restricting it to the labels g_i gives the endpoint $D_- = (w_{g_i})_{i=1}^s$ of a path of weight qm . Indeed, each counted factor acts by a coordinate permutation, so it preserves the corresponding distances; diagonal factors stay diagonal.

For each i , put $r_i = w_{f_i} w_{g_i}^{-1}$. By (26), $r_i \in L$ and $c(r_i) \leq \delta m$. The i th component of $D_+^{-1} D_-$ is

$$w_{f_i}^{-1} w_{g_i} = w_{f_i}^{-1} r_i^{-1} w_{f_i},$$

so it too belongs to L and has cost at most δm . Lemma 6.2 supplies represented $u, v \in M$ such that

$$D_+^{-1} D_- = uv^{-1}, \quad \deg u + \deg v \leq s\delta m.$$

Thus $D_+ u = D_- v$. Writing $a = \deg u$ and $b = \deg v$, the two resulting paths have weights

$$\lambda = m + a > 0, \quad \mu = qm + b \geq 0, \quad \frac{\mu}{\lambda} \leq \frac{qm + b}{m} \leq q + s\delta.$$

□

Completion of the sufficiency implication in Theorem 1.1. Assume the matrix condition of the theorem. By Proposition 2.1, the tensor condition (3) holds. For every integer $n \geq 1$, Lemma 4.1 supplies two images as in Proposition 6.4 with $0 < q < 1/(2n)$. Apply that proposition with $\delta = 1/(2sn)$. It gives equal-endpoint paths whose weight ratio is strictly less than $1/n$. Proposition 5.1 now implies that A is Ramsey. All choices are made after fixing n ; no bound uniform in n on the degrees or the number of coordinates is required. □

7 Geometric consequences and examples

The criterion separates two different questions about a spherical set: whether its ordinary quadratic equations admit the required identity block, and whether that solution lifts to the tensor ring. We first give positive examples, then exhibit two failures of the lift.

Corollary 7.1. *Every finite Euclidean Ramsey set is spherical.*

Proof. A singleton is spherical. Otherwise use the notation and a matrix P from Theorem 1.1, and put $H = m_F(P)$, entrywise. Multiplying the evaluation equations gives

$$\|a_i\|^2 + \ell \cdot a_i + c = 0, \quad c = H_{00}, \quad \ell_\alpha = H_{0\alpha} + H_{\alpha 0} \quad (1 \leq \alpha \leq d).$$

Thus all points lie on the sphere with center $-\ell/2$ and squared radius $\|\ell\|^2/4 - c$. The radius is positive because the set has at least two distinct points. No symmetry of the constant row and column of H is needed. $\square$

A finite Euclidean set is *transitive* if its distance-preserving permutations act transitively on its points.

Corollary 7.2. *Every nonempty subset of a finite transitive Euclidean set is Ramsey.*

Proof. Singletons are immediate. Let $A = \{a_1, \dots, a_s\} \subset \mathbb{R}^d$, with $s \geq 2$, affinely span $\mathbb{R}^d$, and suppose an isometric embedding maps a_i to y_i in a finite transitive set $Y \subset \mathbb{R}^n$. Write H for the finite group of distance-preserving permutations of Y . For each $h \in H$, the map $a_i \mapsto h(y_i)$ extends to an affine isometry. Let $w_{hj} \in W = \mathbb{R}^{1+d}$ be its j th affine coordinate row, so that $e_i(w_{hj}) = (h(y_i))_j$. The gradient Gram matrix of these n rows is I_d .

Let $c_{hj} = ((h(y_1))_j, 0) \in W$, a constant affine row. The symmetric rational tensor

$$T = \frac{1}{|H|} \sum_{h \in H} \sum_{j=1}^n (w_{hj} \otimes w_{hj} - c_{hj} \otimes c_{hj})$$

has $G(T) = I_d$. For each i , the multiset $(h(y_i))_{h \in H}$ contains every point of Y exactly $|H|/|Y|$ times. Consequently

$$\sum_{h,j} (h(y_i))_j \otimes (h(y_i))_j = \sum_{h,j} (h(y_1))_j \otimes (h(y_1))_j,$$

and $(e_i \otimes e_i)(T) = 0$. Proposition 2.1 and Theorem 1.1 now give the Ramsey property. $\square$

Quadratic interpolation gives another source of tensor certificates. The row-independence hypothesis below says that quadratic polynomials can prescribe arbitrary values on the configuration.

Proposition 7.3. *Let $A = \{a_1, \dots, a_s\} \subset \mathbb{R}^d$ be spherical and affinely span $\mathbb{R}^d$. Put $p_i = (1, a_i)^\top$ and let F be its coordinate field. If the s row vectors*

$$(p_{i\alpha} p_{i\beta})_{0 \leq \alpha, \beta \leq d} \quad (1 \leq i \leq s)$$

are linearly independent over F , then A is Ramsey.

Proof. The singleton case is immediate, so assume $s \geq 2$. Write $N = (d+1)^2$. Sphericity gives a real matrix with spatial block I_d and quadratic values zero at every p_i . These requirements are a linear system over F , so a solution exists over F as well, by Gaussian elimination. Denote such a matrix, written as a vector in F^N , by h .

Let $D \in \text{Mat}_{s \times N}(B)$ be the tensor evaluation matrix, whose entry in row i and column (α, β) is $p_{i\alpha} \otimes p_{i\beta}$. Lift h to $b \in B^N$ by applying $x \mapsto x \otimes 1$ entrywise. The assumed row independence allows us to select s columns of D whose square submatrix M has

$$\delta = m_F(\det M) \neq 0.$$

Let $\iota : B^s \rightarrow B^N$ insert a vector into these selected coordinates, so that $D\iota = M$. Multiplying by $\det M$ allows us to cancel the tensor evaluation error Db through the selected columns: the adjugate identity gives

$$z = (\det M)b - \iota \operatorname{adj}(M)Db \in \ker D.$$

Since $m_F(D)h = 0$, multiplication gives $m_F(z) = \delta h$. Multiplying z by $\delta^{-1} \otimes 1$ therefore produces a solution of the criterion in Theorem 1.1. This argument does not require $\det M$ to be a unit of B . $\square$

For at most five circle points, products of two line equations supply the required interpolation. The same observation appears in Pálvölgyi's analysis of derivation obstructions [20, proof of Theorem A.2].

Corollary 7.4. *Every nonempty set of at most five distinct points on a circle is Ramsey. In particular, every cyclic quadrilateral is Ramsey.*

Proof. Work in the plane of the circle and extend the set, if necessary, to five distinct points on that circle. Let F be the coordinate field of the resulting five-point set. Fix one of these points a_i , partition the other four into two pairs, and take the product of affine equations for the two lines through those pairs. No three circle points are collinear, so this quadratic polynomial vanishes at the other four points and is nonzero at a_i . Its coefficients lie in the coordinate field F . Rescaling the five resulting polynomials shows that quadratic evaluation onto F^5 is surjective. Its five rows are therefore independent, and Proposition 7.3 makes the five-point set Ramsey. A monochromatic copy of that set contains a copy of the original set. $\square$

Corollary 7.5. *For every transcendental $a \in (-1, 1)$, the cyclic quadrilateral*

$$K_a = \{(-1, 0), (1, 0), (a, \sqrt{1-a^2}), (a, -\sqrt{1-a^2})\}$$

is Ramsey and is not subtransitive.

Proof. The four points are distinct and lie on the unit circle, so Corollary 7.4 makes K_a Ramsey. Leader, Russell and Walters proved that K_a does not embed in any finite transitive Euclidean set [15, Corollary 2]. $\square$

Thus the necessity direction of the Leader–Russell–Walters subtransitive characterization fails [16, Conjecture A]. The same family also disproves their conjecture that these kites are not Ramsey [15, Conjecture 3].

7.1 Nine points on a circle

The tensor equations can have full rank even when their multiplied versions admit a sphere equation. The following determinant calculation illustrates this distinction. Pálvölgyi's earlier generic-seven-point claim [20, Theorem A.3] already implies its non-Ramsey conclusion by taking a subset; the source labels its appendix as unchecked. We give a direct proof from the tensor criterion.

Proposition 7.6. *Let $t_1, \dots, t_9$ be algebraically independent real numbers over $\mathbb{Q}$. The nine points*

$$a_i = \left(\frac{1-t_i^2}{1+t_i^2}, \frac{2t_i}{1+t_i^2} \right), \quad 1 \leq i \leq 9,$$

form a spherical set that is not Ramsey.

Proof. The parametrization lies on the unit circle and is injective, since $t_i = (a_i)_2 / (1 + (a_i)_1)$. Thus the points are distinct, affinely span $\mathbb{R}^2$, and have coordinate field $F = \mathbb{Q}(t_1, \dots, t_9)$. Algebraic independence identifies this field with a rational function field. Consequently $B = F \otimes_{\mathbb{Q}} F$ is the localization of $\mathbb{Q}[T_1, \dots, T_9, U_1, \dots, U_9]$ obtained by inverting all nonzero polynomials in the T variables alone and all nonzero polynomials in the U variables alone. In particular, B is a domain and embeds in $\mathbb{Q}(T_1, \dots, T_9, U_1, \dots, U_9)$.

Put

$$p(t) = \left(1, \frac{1-t^2}{1+t^2}, \frac{2t}{1+t^2} \right)^T.$$

Over this fraction field, the nine tensor evaluation equations have coefficient matrix D with rows $(p_\alpha(T_i)p_\beta(U_i))_{0 \leq \alpha, \beta \leq 2}$. We claim that $\det D \neq 0$. To check the claim, specialize the nine pairs (T_i, U_i) to the grid $\{0, 1, -1\}^2$, with both coordinates ordered as $0, 1, -1$. All denominators in the displayed matrix remain nonzero. The resulting matrix is the ordinary Kronecker product $C \otimes C$, where

$$C = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & -1 \end{pmatrix}, \quad \det C = 2.$$

Its determinant is $2^6 = 64$, proving the claim. This evaluates the specific determinant expression; no specialization of the entire localized ring B is asserted.

Thus the only solution of $D \operatorname{vec}(P) = 0$ over the fraction field is $P = 0$. Injectivity of the embedding of B shows that the same holds over B . Such a matrix cannot have multiplied spatial block I_2 , so Theorem 1.1 excludes the Ramsey property. $\square$

7.2 Three concentric squares

A second example uses only one transcendental parameter. It also shows how a derivation can detect the failure of the tensor equations directly inside the coordinate field. Its weighted cancellation is related to the derivation obstruction in Pálvölgyi's smaller circular example [20, Theorem 1]; the calculation below is independent of that proof. The parameter is Liouville's constant [17].

Proposition 7.7. *Set*

$$t = \sum_{m=1}^{\infty} 10^{-m!}, \quad u = \frac{1-t^2}{1+t^2}, \quad v = \frac{2t}{1+t^2}, \quad Q_0 = \{(\pm 1, 0), (0, \pm 1)\},$$

and define

$$R_+ = \begin{pmatrix} u & -v \\ v & u \end{pmatrix}, \quad R_- = \begin{pmatrix} u & v \\ -v & u \end{pmatrix}, \quad X = R_+Q_0 \cup R_-Q_0 \cup Q_0.$$

Then X is a twelve-point subset of the unit circle that is not Ramsey.

Proof. First, $0.11 < t < 0.12$ and t is transcendental. Indeed, the rational partial sums $t_m = p_m/q_m$, with $q_m = 10^{m!}$, satisfy

$$0 < t - t_m < 2q_m^{-(m+1)} \quad (m \geq 2).$$

If t had an irreducible polynomial $P \in \mathbb{Z}[T]$ of degree d , then $P(t_m) \neq 0$ and $|P(t_m)| \geq q_m^{-d}$. A bound $C \geq 1$ for $|P'|$ on $[0, 1]$ would give $q_m^{-d} < 2Cq_m^{-(m+1)}$, a contradiction for large m . The rotations R_+ and R_- have angles $\theta = 2 \arctan t$ and $-\theta$, where $0 < \theta < \pi/4$. Their

differences from each other and from zero are nonzero and smaller than $\pi/2$ in absolute value. Hence the three squares are disjoint. They lie on the unit circle, and Q_0 already affinely spans the plane.

The coordinate field of X is $F = \mathbb{Q}(t)$: its coordinates include u, v , and $t = v/(1 + u)$. On this field define the derivation

$$\partial = \frac{1 + t^2}{2} \frac{d}{dt}.$$

It satisfies $\partial u = -v$ and $\partial v = u$. Write $R_0 = I_2$, label the points by $x_{\epsilon, q} = R_{\epsilon} q$ for $\epsilon \in \{+, -, 0\}$ and $q \in Q_0$, and assign weights $\beta_{+, q} = \beta_{-, q} = 1$, $\beta_{0, q} = -2$. For these twelve labels put $p_i = (1, x_i)^\top$. We have the three augmented moment identities

$$\sum_i \beta_i p_i p_i^\top = 0, \quad \sum_i \beta_i (\partial p_i) p_i^\top = 0, \quad \sum_i \beta_i (\partial p_i) (\partial p_i)^\top = \begin{pmatrix} 0 & 0 \\ 0 & 4I_2 \end{pmatrix}. \quad (31)$$

Here the block sizes are 1 and 2. To verify them, use $\sum_{q \in Q_0} q = 0$ and $\sum_{q \in Q_0} qq^\top = 2I_2$. With $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, the derivatives are $\partial R_+ = R_+ J$, $\partial R_- = -R_- J$, and $\partial R_0 = 0$. The first spatial moment is $2(1 + 1 - 2)I_2 = 0$; the mixed moment is $2J - 2J = 0$; and the derivative moment is $2I_2 + 2I_2 = 4I_2$. The constant and vector entries vanish by the same square symmetries.

Define the $\mathbb{Q}$ -linear map $\Phi : F \otimes_{\mathbb{Q}} F \rightarrow F$ by $\Phi(a \otimes b) = (\partial a)(\partial b)$. For every $P \in \text{Mat}_3(B)$, the product rule and (31) give

$$\sum_i \beta_i \Phi((p_i \otimes 1)^\top P (1 \otimes p_i)) = 4 \sum_{\alpha=1}^2 m_F(P_{\alpha\alpha}). \quad (32)$$

For completeness, expand each entry of P into pure tensors $a \otimes b$. In $\partial(p_{i\alpha} a) \partial(p_{i\beta} b)$, the terms involving ∂a or ∂b vanish after summing with β_i by the first two moment identities and the transpose of the second. The remaining term is $ab(\partial p_{i\alpha})(\partial p_{i\beta})$, and the third identity gives exactly the right side of (32).

A matrix satisfying the classification criterion would make the left side zero and the right side 8. This contradiction proves the claim. The argument uses a derivation only on F , and imposes no regularity condition on the colorings in the Ramsey definition. $\square$

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